

Hyperloop Development Program

Reporting Period 1: Nov 2020-Jun 2021

01-09-2021

V1.1

This report highlights the progress of the Hyperloop Development Program. First section summarizes the highlights within the growing hyperloop market, the second section provides an overall program overview, followed by section of progress per working group, and summarized in the financial overview.

REPORTING ON HIGHLIGHTS WITHIN HYPERLOOP MARKET

The past 8 months have shown substantial highlights in the Global developments in the Hyperloop Market, increasing market maturity:

- Virgin Hyperloop demonstrated in **Nevada (US)** the first passengers traveled safely on a hyperloop (08-11-2020): [Link](#)
- Hyperloop Technology Included in **European Union** Sustainable and Smart Mobility Strategy (09-12-2020): [Link](#)
- The **U.S.** Department of Transportation (USDOT) released "Hyperloop Standards Desk Review," a document that provides a preliminary mapping of existing standards and regulations to specific hyperloop systems components (14-01-2021): [Link](#)
- The North Central **Texas (US)** Council of Governments has concluded the first phase of their investigation of potential options for a high-speed link between Dallas and Fort Worth. The next phase will only continue with investigating high-speed rail and hyperloop. (17-05-2021): [Link](#)
- **Canadian** hyperloop company TransPod has completed a feasibility study of the implementation of an ultra-high-speed transportation line between Calgary and Edmonton (25-06-2021): [Link](#)
- Nevomo to build a new test track in **Poland** (01-06-2021): [link](#).
- Results of a hyperloop study in **Flanders (Belgium)** have been made public, with a follow-up to further investigate the market interest into hyperloop developments (10-06-2021): [Link](#)
- HyperloopTT and HHLA (**Hamburg, Germany**) present HyperPort at ITS World Congress (06-07-2021): [Link](#)

HDP has been in the news five times with interviews, podcasts and based on milestones:

- Covenant signed for Hyperloop for Cargo (21-01-2021) - [Link](#)
- The hyperloop may change more than just travel video on Bloomberg (20-05-2021) - [Link](#)
- Podcast BNR Mobility (07-06-2021) - [Link](#)
- EHC event in Groningen (12-06-2021) - [Link](#)
- Interview at nu.nl (12-07-2021) - [Link](#)

HDP has focused on four grants since January 2021 to raise additional funding:

- Growth Fund project proposal for round 2 has been applied for: [Link](#)
- NWA: Proposal to be handed in Q3/Q4 2021, led by TU Delft.
- Interreg North-Sea Region: In the early stages seeking partners for the project, proposal Early 2022.
- Corona R&D Mobiliteitsfonds: stopped due to insufficient match with grant requirements.

REPORTING ON OVERALL PROGRAM PROGRESS

This report is to confirm costs incurred and progress in the Hyperloop Development Program over the period Nov 2020-Jun 2021¹.

The key program deliverable and key deliverable for this stage per working group are represented below.

WG	Key Program deliverable	Key deliverable Concept Stage
WG Cargo	Demonstrated hyperloop for cargo system (TRL-7) in the EHC	Concept Design Cargo approved
WG EHC	European Hyperloop Center (EHC) Phase A completed	Conceptual design EHC Phase A approved & Permits applied
WG FP	Identify and develop 5 routes up to a preliminary level, of which 1 with a go-ahead build-out of route	Final Market opportunity attractiveness
WG Passengers	Passenger hyperloop system concept (TRL-3 & SRL-5)	Final report on the passenger concept review
WG Program Management	Establish a connected hyperloop ecosystem	Consortium Agreement HDP signed

The first few months of the reporting period were focused on preparing the Working Groups and plans to carry out the work in the HDP. The overall HDP program encountered delay in expenditure, as can be seen in the financial section of this report. Consortium Agreement has seen delay, and is now scheduled to be signed by summer 2021 (initial schedule to be signed end of March 2021).

HDP will be finalizing Stage 1 (Nov 2020 – Jun 2021) of a total of four stages in summer 2021. The program milestones (marked in red in the stream “Program Management”) will be evaluated on progress and deliverables of each working group before progressing to the next stage.

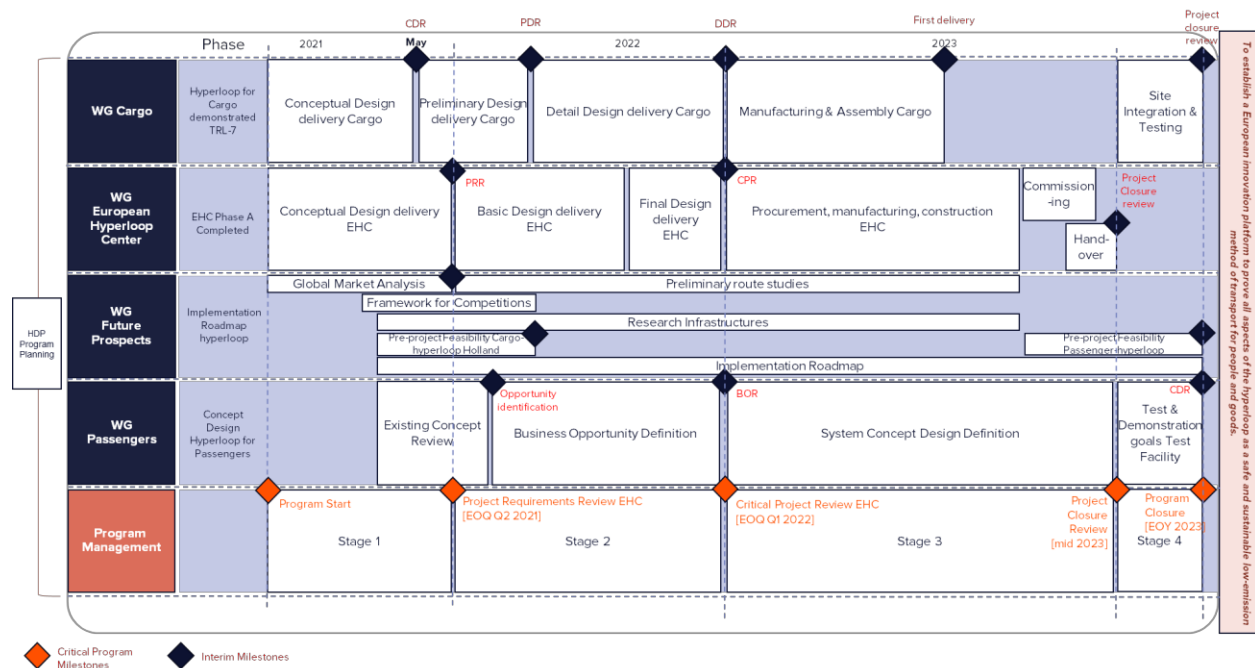


Figure 1: Stages & Milestone planning HDP

¹ The report extends the earlier report from 11-05-2021: *Aanvraag subsidievoorschot Hyperloop Development Program*.

The Program deliverable status for stage 1 is presented below. A number of milestones have encountered delays, and will be completed in September 2021, instead of EOQ Q2.

#	Milestone	Working Group	Status	Date delivery
1	Concept Design Cargo approved	Cargo	Completed	-
2	Test Plan EHC defined	Cargo	Completed	-
3	Preliminary outline EHC Phase A established	Cargo + EHC	Completed	-
4	Plan & budgets Preliminary Design approved	Cargo	On-going	Sept 21
5	Available of Program Funding & Resources secured for EHC (2), (3)	PM + EHC	On-going	Sept 21
6	Phasing of EHC with Program funding towards complete scope (2,6 km) established	EHC	Completed	-
7	Conceptual design EHC Phase A approved & Permits applied	EHC	Delayed	Q2-22
8	Plan & budgets Basic Design approved	EHC	Delayed	Sept 21
9	Final Market opportunity attractiveness	Future Prospects	Delayed	Sept 21
10	Alignment preliminary market needs/wishes/requirements of Pre-project feasibility Cargo	Future Prospects	Completed	-
11	Established plan and process to execute the 5 preliminary route studies	Future Prospects	On-going	Sept 21
12	First outlook for Roadmap NL has been established	Future Prospects	Delayed	Sept 21
13	Framework for competitions finished	Future Prospects	Completed	-
14	Final report on the passenger concept review	Passenger	On-going	Aug 21
15	Check status partners & Stakeholders Green light	Program Management	On-going	-
16	Consortium Agreement HDP signed	Program Management	Delayed	Sept 21
17	Plans approved and funding secured for stage 2	Program Management	On-going	Sept 21

Changes in the Work Packages

- The Work Packages Pre-project Feasibility Cargo (WG Cargo) and Pre-project Feasibility Passengers (WG Passengers) have been relocated to WG Future Prospects.²

Changes in the Ecosystem

Partners onboarded, of which two will not make any eligible cost for the subsidy.³

Partner	Working Group / WP	Eligible Cost	Subsidy
Berenschot	Future Prospects: Hyperloop Implementation Roadmap	€ 67.963	€ 16.311 (24%)
TUV Rheinland	Cargo	€ 0	€ 0
Dutch Boosting Group	n/a	€ 0	€ 0

Partners exit:

- Balance. The eligible costs of Balance in the WG Cargo (€ 413.030) has been marked as "Partner TBD".⁴

² Has been approved by the Grant provider in letter 'voorschot subsidie verplichtingnummer 1300031782' dated 15 June 2021.

³ Has been approved by the Grant provider in letter 'voorschot subsidie verplichtingnummer 1300031782' dated 15 June 2021.

⁴ Has been approved by the Grant provider in letter 'voorschot subsidie verplichtingnummer 1300031782' dated 15 June 2021.

REPORTING PER WORKING GROUP

WG Cargo

Status: Proceeding according to WG Plan

The Working Group Cargo established the Technical Management document. The Working Group delivered the concept design of the hyperloop for Cargo.

The team has finished a number of deliverables:

- on *system design*: Prototypes, Operational Control Concept, Propulsion system concept, primary suspension concept.
- on *public adoption*: Cargo service vision, product vision, business case, stakeholder needs,
- on *safety* and *standardization* the following deliverables have been delivered: Functional Hazard Analysis, Safety vision, Safety Concept.
- in conjunction with WG EHC, the scope of EHC Phase A has been established (Project Brief).

In the period the standardization National Committee has convened twice in January and March 2021, and the JTC-20 (EU) has convened twice on February and April 2021. The JTC-20 has been structured into 2 work groups, and 5 work items ([link](#)). Participants at the National Committee are Hardt, TataSteel, Movares and Dutch Boosting Group.

Next key milestone:

- Q4 2021: Delivery Preliminary Design

WG Passengers

Status: Delayed start to baseline planning

The Working Group started with evaluation of existing hyperloop concept for passengers, conducted by ADSE. Scoping meetings with NS have taken place to start the work on the Operational Control Concept. Further scoping meetings with other partners will take place in September.

Deliverables finished:

- Final report on the passenger concept review delivered

Next key milestone:

- Dec 2021: Delivery of Operational Control Concept
- Q2 2022: Delivery Business Opportunity Definition

WG EHC

Status: Delayed to baseline planning

The Working Group EHC (Stichting EHC) has been primarily focused on project planning of the EHC Phase A, by delivery of Project Plan, Project Management Plan. In conjunction with WG Cargo, the scope of EHC Phase A has been established (Project Brief). The “Environmental Vision” (*Gebiedsvisie*) and EIA-assessment (*MER-beoordeling*) have been approved. The team is now preparing official permit application for *Omgevingsvergunning Ruimtelijke Ordening*.

Further the team has:

- finalized a number of studies (archaeological, ecological)
- developed a spatial integrated design of the EHC in the area
- organized three sessions with the local residents to inform on the progress

Deliverables finished:

- Project Plan, Project Management Plan, Project Brief

Next key milestones:

- 2022 Q2 permit approval *Omgevingsvergunning Ruimtelijke Ordening*

WG Future Prospects

Status: Proceeding according to WG Plan

The working group has started its activities on Work Packages Global Market Analysis, Preliminary Route Studies, Pre-Project Feasibility Cargo and Research Infrastructures. A more detailed WG Plan has been established and will form the basis for the further work. The activities for Implementation Roadmap are delayed as the onboarding respectively acquisition of new partners took longer than anticipated.

The following things need to be highlighted:

- WP Pre-Project Feasibility Cargo has been shifted from WG Cargo to WG Future Prospects
- WP Pre-Project Feasibility Passenger has been shifted from WG Passenger to WG Future Prospects
- Berenschot has been added as a new partner within this WG for Work Package Implementation Roadmap focusing on activity “Implementation Roadmap Netherlands”

Deliverables finished:

- WG Plan (WP 0), Business Case Report (WP 1),
- Framework for competitions (WP 6)

Next key milestones:

- August 2021: Report on global market opportunities (WP 1 – Global Market Assessment)
- August 2021: Outlook on hyperloop in Netherlands (WP 7 – Hyperloop Implementation Roadmap)
- Q3 2021: Delivery Pre-project feasibility hyperloop for Cargo